

STATISTICS 652

LECTURE 6 - 26

REVIEW OF LECTURE 5 AND SOME CONCEPTS FROM STAT 593

For $\underline{X} \sim F_\theta$, $\theta \in (H) \subset \mathbb{R}$, THE INFORMATION FOR $\underline{\theta}$ IN $\underline{X}$ IS

$$I(\theta; \underline{X}) = E\left[\left\{2 \log f(\underline{X}; \theta) / 2\theta\right\}^2\right] \quad (1)$$

$$= \text{VAR}\left[2 \log f(\underline{X}; \theta) / 2\theta\right] \quad (2)$$

$$= -E_\theta\left[2^2 \log f(\underline{X}; \theta) / 2\theta^2\right] \quad (3)$$

$$\textcircled{1} \Rightarrow \textcircled{2} \quad \text{SINCE} \quad E\left[2 \log f(\underline{X}; \theta) / 2\theta\right] = 0$$

PROOF: $1 = \int f(\underline{x}; \theta) d\underline{x}$, TAKING DERIVATIVE ON BOTH SIDES

AND UNDER REGULARITY CONDITIONS WHICH ALLOW INTERCHANGE OF DIFFERENTIATION AND INTEGRATION GIVES

$$\begin{aligned} 0 &= \frac{\partial}{\partial \theta} \int f(\underline{x}; \theta) d\underline{x} = \int \left\{ \frac{\partial}{\partial \theta} f(\underline{x}; \theta) / f(\underline{x}; \theta) \right\} f(\underline{x}; \theta) d\underline{x} \\ &= \int \left\{ 2 \log f(\underline{x}; \theta) / 2\theta \right\} f(\underline{x}; \theta) d\underline{x} = E_\theta\left[2 \log f(\underline{X}; \theta) / 2\theta\right]. \end{aligned}$$

$$\textcircled{1} \Rightarrow \textcircled{3} \quad \text{SINCE}$$

$$\frac{2 \log f(\underline{X}; \theta)}{2\theta^2} = - \left\{ \frac{2 \log f(\underline{X}; \theta)}{2\theta} \right\}^2 + \frac{2 f(\underline{X}; \theta) / 2\theta^2}{f(\underline{X}; \theta)}$$

with the expected value of second term

being 0. VERIFY BY TWICE DIFFERENTIATING $1 = \int f(\underline{x}; \theta) d\underline{x}$

RECALL: FOR $\psi(\theta) = E_{\theta}[H(\underline{X})]$, $\psi'(\theta) = \text{cov}_{\theta}(H(\underline{X}), \frac{\partial \log f(\underline{X}; \theta)}{\partial \theta})$

PROOF: $\psi(\theta) = \int H(\underline{x}) f(\underline{x}; \theta) d\underline{x} \Rightarrow$

$$\begin{aligned} \psi'(\theta) &= \int H(\underline{x}) \frac{\partial f(\underline{x}; \theta)}{\partial \theta} d\underline{x} = \int H(\underline{x}) \frac{\partial \log f(\underline{x}; \theta)}{\partial \theta} \cdot f(\underline{x}; \theta) d\underline{x} \\ &= E_{\theta} \left[H(\underline{X}) \frac{\partial \log f(\underline{X}; \theta)}{\partial \theta} \right] = \text{cov}_{\theta} \left(H(\underline{X}), \frac{\partial \log f(\underline{X}; \theta)}{\partial \theta} \right) \end{aligned}$$

SINCE $E \left[\frac{\partial \log f(\underline{X}; \theta)}{\partial \theta} \right] = 0$.

FOR ONE PARAMETER EXPONENTIAL FAMILY

$$f(\underline{x}; \theta) = \exp \{ c(\theta) T(\underline{x}) + d(\theta) + s(\underline{x}) \} I_A(\underline{x})$$

OR IN NATURAL FORM

$$f_0(\underline{x}; \eta) = \exp \{ \eta T(\underline{x}) + d_0(\eta) + s(\underline{x}) \} I_A(\underline{x})$$

with $\eta = c(\theta)$ being 1-1 and $d_0(\eta) = d(\theta)$ AND HENCE $d(\theta) = d_0(c(\theta))$

INFORMATION FOR η GIVEN $\underline{X}$: $I_0(\eta; \underline{X}) = -d_0''(\eta)$

INFORMATION FOR θ GIVEN $\underline{X}$:

$$\begin{aligned} I(\theta; \underline{X}) &= I_0(c(\theta); \underline{X}) \{c'(\theta)\}^2 \\ &= -d''(\theta) + d'(\theta) c''(\theta) / c'(\theta) \quad \text{PROBLEM 1.13.} \end{aligned}$$

REVIEW POISSON EXAMPLE:

$X_1, \dots, X_n$ iid Poisson (λ)

$$f(\underline{x}; \lambda) = \exp \{ (\log \lambda) n \bar{x} - n\lambda + S(\lambda) \} I_A(\underline{x})$$

NATURAL PARAMETER: $\eta = \log \lambda$ $d_0(\eta) = -ne^\eta$ with $c(\lambda) = \log \lambda$

$$\Rightarrow I_0(\eta; \underline{x}) = -d_0''(\eta) = ne^\eta$$

$$\Rightarrow I(\underline{x}; \lambda) = I_0(c'(\lambda); \underline{x}) \{c'(\lambda)\}^2 = n\lambda \cdot (1/\lambda)^2 = n/\lambda \quad \text{as BEFORE}$$

NOTE: UTILIZE THE GENERAL RESULTS FOR EXPONENTIAL FAMILIES!

RECALL: $\hat{\gamma} = \left(1 + \frac{1}{n}\right)^{n\bar{x}}$ IS UMVUE FOR $\gamma = e^\lambda$ (HW)

HW: FIND INFORMATION FOR γ IN $\underline{x}$.

NOTE: MLE OF γ IS $\hat{\gamma} = e^{\bar{x}}$ SINCE $\bar{x}$ IS MLE OF λ .

"CLOSE" TO UMVUE SINCE $\left(1 + \frac{1}{n}\right)^n \rightarrow e$ AS $n \rightarrow \infty$.

NOTE: REGULARITY CONDITIONS FOR INFORMATION AND FOR ASYMPTOTIC NORMALITY OF MLEs HOLD FOR FULL RANK EXPONENTIAL FAMILIES.

RECALL: ASYMPTOTIC NORMALITY OF MLE

SUPPOSE $X_1, \dots, X_n$ REPRESENTS A RANDOM SAMPLE OF $X \sim F$ WHERE

$F \in \mathcal{F} = \{F_\theta : \theta \in (H)\}$, $(H) \subset \mathbb{R}$. LET $\hat{\theta}_n$ REPRESENT THE

MAXIMUM LIKELIHOOD ESTIMATE OF θ AND $g: \mathbb{R} \rightarrow \mathbb{R}$ BEING DIFFERENTIABLE. THEN UNDER CERTAIN REGULARITY CONDITIONS

$$\sqrt{n} (g(\hat{\theta}_n) - g(\theta)) \rightarrow_d N(0, \sigma_{\theta, g}^2)$$

where $\sigma_{\theta, g}^2 = \frac{\{g'(\theta)\}^2}{I(\theta; \underline{x})}$ AND $\sigma_{\theta, g}^2 = \frac{1}{I(g(\theta); \underline{x})}$ when g is 1-1.

SKETCH OF PROOF,

- CONSISTENCY: Let $l(\theta; \underline{X}) = \log f(\underline{X}; \theta)$. THEN BY S.L.L.N,

$$\frac{1}{n} \sum_{i=1}^n l(\theta; \underline{X}_i) \xrightarrow{\text{almost surely}} E_{\theta_0}(l(\theta; \underline{X})) \quad \text{when } \underline{X} \sim F_{\theta_0} \text{ for each fixed } \theta \in \Theta$$

$$\Rightarrow \hat{\theta}_n \xrightarrow[\text{a.s.}]{\text{ARG SUP}_{\theta \in \Theta}} E_{\theta_0}(l(\theta; \underline{X})) = \theta_0 \quad \text{under regularity condition (ADVANCED PROOF)}$$

- SHOW: $\theta_0 = \text{ARG SUP}_{\theta \in \Theta} E_{\theta_0}(l(\theta; \underline{X}))$

Since $\log$ is a CONCAVE function, by JENSEN'S INEQUALITY

$$\begin{aligned} E_{\theta_0}[l(\theta; \underline{X})] - E_{\theta_0}[l(\theta_0; \underline{X})] &= E_{\theta_0}[\log \{f(\underline{X}; \theta) / f(\underline{X}; \theta_0)\}] \\ &\leq \log E_{\theta_0}[f(\underline{X}; \theta) / f(\underline{X}; \theta_0)] = \log \int f(\underline{X}; \theta) dN = \log 1 = 0. \end{aligned}$$

$\Rightarrow \theta_0$ MAXIMIZE $E_{\theta_0}[l(\theta; \underline{X})]$ OVER $\theta \in \Theta$. ALSO, IT IS UNIQUE MAXIMUM SINCE INEQUALITY IS STRICT UNLESS $f(\underline{X}; \theta) = f(\underline{X}; \theta_0)$.

- ESTIMATING EQUATION: Let $\psi(\underline{X}; \theta) = \frac{\partial l(\theta; \underline{X})}{\partial \theta} = \frac{\frac{\partial f(\underline{X}; \theta)}{\partial \theta}}{f(\underline{X}; \theta)}$

$$\text{THEN } 0 = \sum_{i=1}^n \psi(\underline{X}_i; \hat{\theta}_n) \quad \text{AND} \quad 0 = E_{\theta_0}[\psi(\underline{X}; \theta_0)]$$

$$\begin{aligned} \Rightarrow 0 &= \sum_{i=1}^n \left\{ \psi(\underline{X}_i; \theta_0) + (\hat{\theta}_n - \theta_0) \frac{\partial \psi(\underline{X}_i; \theta)}{\partial \theta} \Big|_{\theta=\theta_0} \right. \\ &\quad \left. + \frac{1}{2} (\hat{\theta}_n - \theta_0)^2 \frac{\partial^2 \psi(\underline{X}_i; \theta)}{\partial \theta^2} \Big|_{\theta=\theta_*} \right\} \end{aligned}$$

where $\theta_* \in [\hat{\theta}_n, \theta_0]$

so

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \frac{-\sqrt{n} \left\{ \frac{1}{n} \sum_{i=1}^n \psi(X_i; \theta_0) \right\}}{\frac{1}{n} \sum_{i=1}^n \frac{2\psi(X_i; \theta)}{2\theta} \bigg|_{\theta=\theta_0} + \frac{1}{2} (\hat{\theta}_n - \theta_0) \frac{1}{n} \sum_{i=1}^n \frac{2\psi^2(X_i; \theta)}{2\theta^2} \bigg|_{\theta=\theta_0}}$$

now

1) By SLN: $\frac{1}{n} \sum_{i=1}^n \frac{2\psi(X_i; \theta)}{2\theta} \bigg|_{\theta=\theta_0} \xrightarrow{as} g(\theta_0) = E_{\theta_0} \left[\frac{2\psi(X; \theta)}{2\theta} \right] \bigg|_{\theta=\theta_0}$

2) By consistency $\hat{\theta}_n - \theta_0 \xrightarrow{as} 0$, and by SLN

$$\frac{1}{n} \sum_{i=1}^n \frac{2\psi^2(X_i; \theta)}{2\theta^2} \bigg|_{\theta=\theta_0} = O_p(1), \text{ hence second term in denominator } \rightarrow 0.$$

3) By CLT and since $E_{\theta_0}[\psi(X; \theta_0)] = 0$,

$$\sqrt{n} \left\{ \frac{1}{n} \sum_{i=1}^n \psi(X_i; \theta_0) \right\} \xrightarrow{d} Z \sim N(0, E_{\theta_0}[\psi^2(X; \theta_0)])$$

4) By Slutsky's Theorem:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \frac{Z}{g(\theta_0)} \sim N(0, \sigma_{\theta_0}^2 = \frac{E_{\theta_0}[\psi^2(X; \theta_0)]}{E_{\theta_0} \left[\frac{2\psi(X; \theta)}{2\theta} \right]^2 \bigg|_{\theta=\theta_0}})$$

HW: SHOW: $\sigma_{\theta}^2 = 1/I(\theta; X)$ (NOTE: $I(\theta; X)/I^2(\theta; X)$)

HW: BY DELTA METHOD, SHOW RESULT FOR $\sqrt{n}(g(\hat{\theta}_n) - g(\theta))$ FOLLOWS,

NOTE: THIS RESULT CAN BE GENERALIZED TO M-ESTIMATES, (VAN DER VAART, 1996)

LECTURE 6 - PART 2 OR LATER

MULTIPARAMETER PROBLEMS.

REVIEW (?): MULTIVARIATE NORMAL DISTRIBUTION

For $X \in \mathbb{R}^p$, $X \sim N_p(\mu, \Sigma)$ with $\mu \in \mathbb{R}^p$ and $\Sigma_{p \times p} > 0$ SYMMETRICDENSITY IN $\mathbb{R}^p$: $f(x; \mu, \Sigma) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left\{-\frac{1}{2}(x-\mu)' \Sigma^{-1} (x-\mu)\right\}$
 $x \in \mathbb{R}^p$

(WHY IS THIS A VALID DENSITY?)

RANDOM SAMPLE: $X_1, X_2, \dots, X_n$ iid $N_p(\mu, \Sigma)$

$$f(x_1, \dots, x_n; \mu, \Sigma) = \exp\left\{-\frac{1}{2} \text{TRACE}\{\Sigma^{-1} T_1\} + \mu' \Sigma^{-1} T_2 + d(\mu, \Sigma)\right\}$$

$$\text{where } T_1(x_1, \dots, x_n) = \sum_{i=1}^n x_i x_i' \quad \text{AND} \quad T_2(x_1, \dots, x_n) = \sum_{i=1}^n x_i$$

VERIFY THIS EXPRESSION, HENCE, EXPONENTIAL FAMILY, FULL RANK

 $\Rightarrow$ NATURAL MINIMAL SUFFICIENT STATISTIC: $\{T_1, T_2\}$ ALSO, $(\bar{X}, S_n)$ ARE MINIMAL SUFFICIENT STATISTICS SINCE

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} T_2$$

ARE 1-1
FUNCTIONS
OF
 (T_1, T_2)

$$\text{and } S_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' = \frac{1}{n} \sum_{i=1}^n X_i X_i' - \bar{X} \bar{X}'$$

$$= \frac{1}{n} T_1 - \frac{1}{n} T_2 T_2'$$

NATURAL PARAMETERIZATION: $\eta = \{\eta_1 = \frac{1}{n} \sum_{i=1}^n x_i, \eta_2 = \frac{1}{n} \sum_{i=1}^n x_i^2\}$

M.L.E: $T_1 = E[T_1] = n(\mu \mu' + \frac{1}{n}) \Rightarrow \begin{cases} \hat{\mu} = \bar{x} \\ \hat{\frac{1}{n}} = S_n \end{cases}$ IS UNIQUE MLE FOR $(\mu, \frac{1}{n})$.

$T_2 = E[T_2] = n \mu$

NOTE: PROVING MLE DIRECTLY IS COMPLICATED. SEE RAD (1973), SEC 8.9.5

UNIQUE UMVUE: EXPRESS $\bar{X} = \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{pmatrix}$, $S = \{S_{ij}\}$, $i, j = 1, \dots, p$ where $S = \frac{n}{n-1} S_n$

$\mu = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix}$, $\frac{1}{n} = \{\sigma_{ij}\}$, $i, j = 1, \dots, p$

BY THEORY OF EXPONENTIAL FAMILIES

$\bar{X}_j$ IS UMVUE FOR μ_j , S_{ij} IS UMVUE FOR σ_{ij}

ALSO: $a' \bar{X}$ IS UMVUE FOR $a' \mu$, $a' S b$ IS UMVUE FOR $a' \frac{1}{n} b$

NOTE: $E[a' \bar{X}] = a' \mu$ $E[a' S b] = a' \frac{1}{n} b$

$E[\bar{X}] = \mu$

$E[S] = \frac{1}{n} \frac{1}{n}$

$VAR(a' \bar{X}) = \frac{1}{n} a' \frac{1}{n} a$

$VAR(a' S b) = \frac{1}{n-1} \{a' \frac{1}{n} a b' \frac{1}{n} b + (a' \frac{1}{n} b)^2\}$

$VAR(\bar{X}) = \frac{1}{n} \frac{1}{n}$

$VAR(S) = ?$ EXPRESSION?

REVIEW OF MULTIVARIATE PROBABILITY AND STATISTICS - LATER

NEXT TIME: SHORT COURSE ON ROBUST STATISTICS